

SINAIA, Romania

WMO: 153250

Lat: 45.3550N

Lon: 25.5142E

Elev: 4958

StdP: 12.25

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	2.9	6.7	-5.9	4.8	13.1	-0.3	6.5	14.8	40.5	15.3	31.5	16.9	4.5	0	0.611

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	13.8	73.8	61.5	71.1	59.6	68.6	58.4	63.2	71.8	61.5	69.2	60.0	67.0	2.3	20

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
60.0	93.2	67.5	58.5	88.1	65.7	57.1	83.7	64.2	31.4	72.3	30.0	69.1	28.8	67.1	68.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
25.8	18.5	14.5	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	41.5	23.2	27.8	30.4	38.7	48.4	56.3	57.6	58.6	50.5	42.3	35.4	27.6
	DBStd	14.86	9.90	8.85	8.75	10.24	6.90	6.46	6.11	6.57	8.86	9.44	9.23	8.97
	HDD50	4088	831	622	609	357	111	19	11	8	106	279	440	695
	HDD65	8620	1296	1042	1074	789	514	268	236	210	440	704	887	1160
	CDD50	973	0	0	0	18	62	207	248	276	119	40	3	0
	CDD65	31	0	0	0	0	0	6	8	13	4	0	0	0
	CDH74	87	0	0	0	0	0	20	23	33	11	0	0	0
	CDH80	4	0	0	0	0	0	1	1	2	0	0	0	0
	WSAvg	4.9	5.5	5.4	5.8	5.3	4.4	4.4	4.4	4.4	4.5	4.7	5.0	5.0
	PrecAvg	34.8	1.7	1.6	2.0	2.8	4.1	4.8	4.5	3.6	2.9	2.6	2.0	2.1
Precipitation	PrecMax	48.9	3.7	3.5	3.8	6.3	8.1	8.0	8.9	9.3	5.6	5.2	4.4	5.6
	PrecMin	21.1	0.4	0.2	0.5	1.3	1.0	1.9	1.5	1.0	1.0	0.2	0.1	0.3
	PrecStd	6.9	0.9	0.7	0.9	1.3	1.9	1.9	1.9	2.0	1.5	1.3	1.1	1.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	49.4	50.0	55.4	64.6	68.6	76.6	76.6	77.5	74.2	66.9	59.3	54.0
		MCWB	36.8	40.7	43.2	50.7	54.5	64.0	62.5	63.0	60.7	52.3	45.9	39.2
	2%	DB	41.9	46.2	50.2	60.7	65.5	71.9	73.2	73.7	69.1	63.4	53.7	46.2
		MCWB	35.6	38.3	40.2	48.5	54.2	60.4	62.0	60.9	56.9	50.5	43.9	36.0
	5%	DB	38.9	42.7	46.2	56.6	62.0	68.7	69.9	70.9	65.2	58.8	50.1	41.8
		MCWB	33.6	35.9	38.2	46.4	52.8	59.3	60.1	59.6	55.0	48.4	42.4	35.2
	10%	DB	35.8	39.6	42.6	52.5	58.7	65.8	66.7	67.7	61.6	54.8	47.0	38.4
		MCWB	32.0	34.1	36.3	44.1	51.0	57.6	58.5	58.5	53.6	46.3	40.3	33.9
	0.4%	WB	38.7	42.9	44.1	52.8	57.3	65.0	64.7	65.3	61.2	55.2	47.9	41.0
		MCDB	43.8	48.2	53.9	61.4	65.8	72.6	74.1	75.5	72.5	63.9	54.6	47.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	36.9	39.4	41.5	49.8	55.0	62.1	62.9	63.0	57.9	52.5	46.0	38.6
		MCDB	40.8	44.6	48.9	59.6	63.0	70.3	71.1	71.5	67.6	60.0	51.1	43.4
	5%	WB	34.9	37.0	39.0	47.6	53.6	60.3	61.2	61.2	56.2	50.2	44.4	36.7
		MCDB	38.5	41.4	45.0	55.4	60.7	67.2	68.3	68.7	64.5	57.4	48.9	40.7
	10%	WB	32.2	35.1	36.9	45.0	51.6	58.6	59.6	59.5	54.4	47.8	42.4	34.7
		MCDB	35.0	39.1	42.0	51.5	57.9	65.2	65.7	66.9	61.1	53.6	46.6	38.2
Mean Daily Temperature Range	5% DB	MDBR	11.6	12.3	12.4	13.6	13.1	14.1	13.9	13.8	12.5	12.1	10.8	10.9
		MCDBR	15.1	15.2	17.2	18.1	17.0	17.6	17.0	16.6	16.5	16.9	14.4	14.0
		MCWBR	10.1	10.1	10.5	11.1	11.4	11.3	11.2	10.4	10.7	12.1	10.0	9.2
	5% WB	MCDBR	13.4	13.6	15.7	16.8	16.5	16.8	16.1	15.9	15.6	16.5	11.2	12.2
		MCWBR	9.5	9.8	10.4	10.5	11.2	11.4	11.2	10.7	10.2	12.2	8.9	8.9
Clear-Sky Solar Irradiance	taub		0.264	0.288	0.322	0.370	0.355	0.357	0.368	0.380	0.344	0.315	0.293	0.267
	taud		2.400	2.349	2.344	2.261	2.353	2.366	2.334	2.298	2.387	2.472	2.489	2.444
	Ebn at Noon		273	285	288	281	289	288	283	275	277	272	259	260
	Edh at Noon		23	29	34	40	38	38	39	38	32	26	21	20
All-Sky Solar Radiation	RadAvg		443	704	1049	1374	1676	1815	1846	1666	1220	838	487	363
	RadStd		48	91	110	150	160	160	154	134	115	107	51	45

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (73 neighbors)	+0.99	N/A	N/A	+1.19	N/A	N/A	-207	-290	+130	+43

Nomenclature: See separate page